What Do Head Scans Reveal About Depression?

Insights from 360° Psychomotor Assessment

Report 1: Descriptive Statistics & Preliminary Analysis

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Course: Behavioral Research & Statistical Methods

1 Introduction

Depression is one of the most prevalent mental health disorders worldwide, affecting over 280 million people [1]. Traditional diagnostic tools, such as the Patient Health Questionnaire-9 (PHQ-9), rely exclusively on self-report, which is susceptible to recall bias, social desirability effects, and limited patient insight [2]. Developing objective behavioural biomarkers for depression has therefore become a significant research priority.

A promising avenue is **psychomotor retardation** the slowing of physical and mental activity that is a core diagnostic criterion of major depressive episodes [3]. Srivastava and Lahane [4] proposed that head movement patterns during immersive virtual reality (VR) exposure could serve as such a biomarker. Using 360° video stimuli, they showed that individuals with moderate-to-severe depression exhibited significantly slower head scanning speeds compared to those with minimal symptoms, even when self-reported emotional ratings were comparable across groups.

The present study is a **replication** of Srivastava and Lahane [4], conducted with a student population at IIIT Hyderabad. Forty participants were exposed to five 360° VR videos spanning different emotional contexts (abandoned buildings, beach, campus, horror, surfing) using a **Meta Quest 3** headset a hardware difference from the original study's HTC Vive Pro. This study added **PANAS** (Positive and Negative Affect Schedule) [5] before and after the VR session, as well as per-video presence/immersion ratings. Code and Data is available at this repo: `https://github.com/hrithiksagar/brsm_s26_project_vr`

2 Methods

2.1 Participants

Forty participants ($M_{\text{age}} = 22.77$, $SD = 1.80$, range 19-27; 36 male, 4 female by assuming gender:2 is for female) from the IIIT Hyderabad community participated.

2.2 Materials and Procedure

Participants watched five 360° videos in fixed order using a Meta Quest 3 headset:

- **V1** Abandoned Buildings

- **V2** Evening at the Beach
- **V3** University Campus
- **V4** The Nun Horror
- **V5** Tahiti Surf

After each video, participants rated **valence** (1-9), **arousal** (1-9), and an **immersion** scale. Questionnaires administered included PHQ-9 [6], GAD-7 [7], and STAI-T [8]. PANAS [5] was collected immediately before and after the VR block. Head orientation was recorded continuously by the Meta Quest 3; the key metric extracted from each recording is **mean yaw angular speed** (RotationSpeedY, deg/s), which measures how actively a participant scans left and right during a video. Yaw was selected as the primary axis because 360° video exploration is predominantly horizontal; pitch (up-down) and roll (tilt) are less informative for scene-scanning behaviour, and translational position changes (X, Y, Z) are negligible in a seated task. This metric directly matches the operationalisation used by Srivastava and Lahane [4].

2.3 Analysis Plan

All analyses were conducted in Python using `scipy`, `pandas`, `matplotlib`, and `seaborn`. Normality of all key variables was assessed using the Shapiro-Wilk test (see Results). Pearson and Spearman correlations were computed among clinical measures and between PHQ-9 and head-tracking data, respectively; Spearman was used for head-tracking correlations given normality violations in those variables. Given that PHQ-9, GAD-7, and STAI-T violated normality, a Kruskal-Wallis test the non-parametric, as used in Srivastava and Lahane [4] (equivalent) was used to compare groups. A paired t-test was applied to PANAS difference scores, which were normally distributed.

3 Results

3.1 Descriptive Statistics and Normality

Table 1 presents descriptive statistics for all key variables. PHQ-9, GAD-7, and STAI-T all violated normality (Shapiro-Wilk $p \leq .045$). PANAS difference scores were approximately normal ($p > .10$), permitting a paired t-test. Figure 1 illustrates the full distributional picture.

Table 1: Descriptive statistics for key variables ($N = 40$).

Measure	Mean	SD	Min	Max	Skewness	SW p
Age	22.77	1.80	19	27	0.42	.091
PHQ-9	6.03	4.63	0	18	0.81	.002**
GAD-7	5.00	4.31	0	18	0.98	<.001**
STAI-T	45.00	14.63	21	73	0.31	.045*
PANAS Positive (Pre)	34.52	7.33	19	50	−0.11	.214
PANAS Positive (Post)	32.15	10.30	10	50	−0.34	.183
PANAS Negative (Pre)	12.70	3.52	10	27	1.62	.003**
PANAS Negative (Post)	13.95	6.80	10	46	2.06	<.001**

Note. SW = Shapiro-Wilk. * $p < .05$, ** $p < .01$.

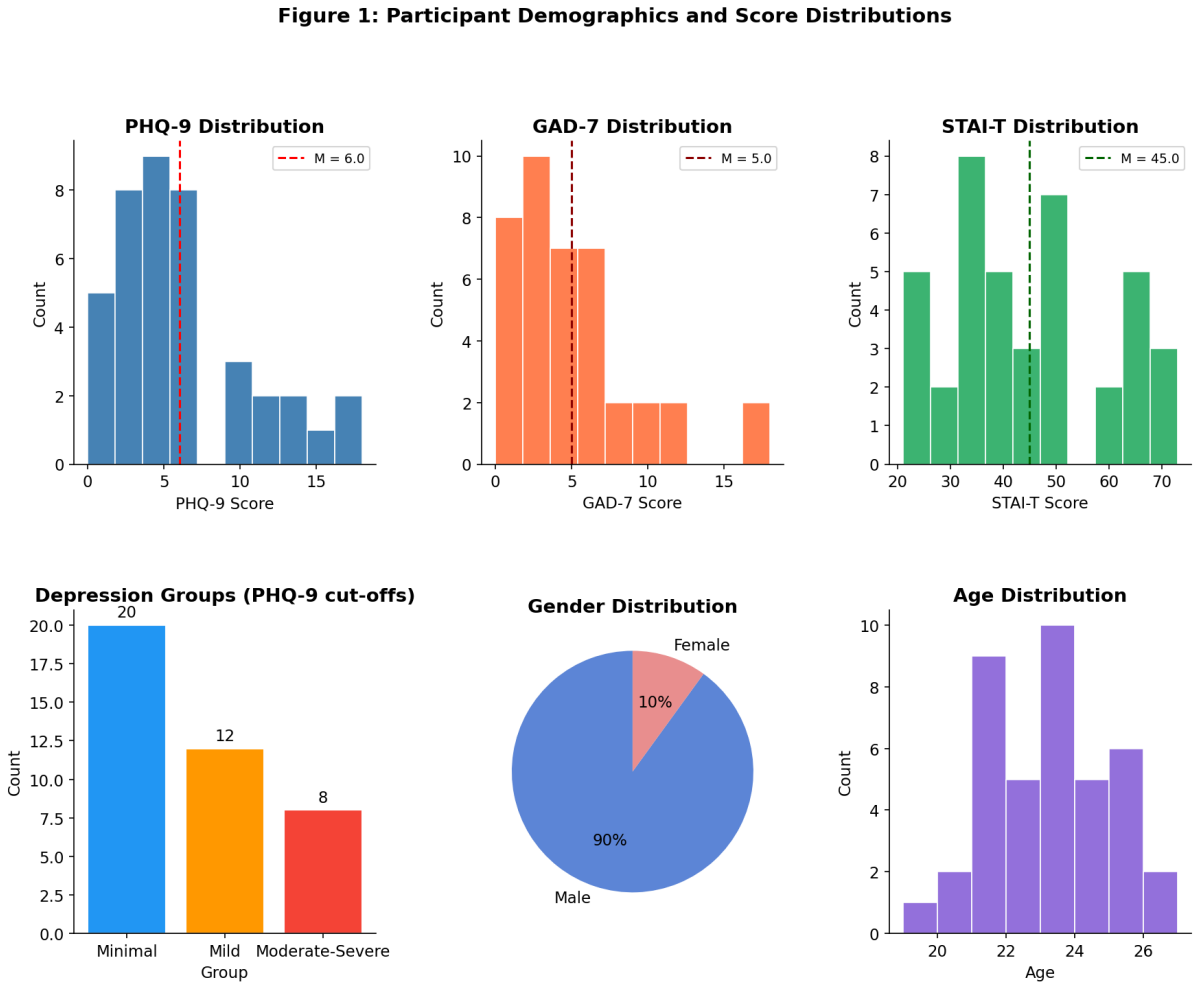

Figure 1: PHQ-9, GAD-7, and STAI-T score distributions; depression group frequencies; gender and age distributions ($N = 40$).

3.2 Outlier Detection and Inclusion Decisions

Outliers were assessed via the IQR method and z-score thresholding ($|z| > 2.5$). PHQ-9 flagged three participants (scores 16–18) by IQR and one (score 18) by z-score; GAD-7 flagged two. All flagged values are clinically valid and represent the Moderate-to-Severe group removing them would eliminate the most important subgroup for the study's hypothesis [9]. All 40 participants were therefore retained.

3.3 Participant Grouping by Depression Severity

Table 2 shows participant distribution using validated PHQ-9 cut-offs [6]: Minimal (0–4), Mild (5–9), Moderate-Severe (≥10). A three-category scheme was used to preserve ordinal information and avoid the power loss associated with binary splits.

Table 2: Participant distribution by PHQ-9 depression severity group.

Group	n	PHQ-9 Range	M (SD)	%
Minimal Depression	20	04	2.60 (1.47)	50%
Mild Depression	12	59	6.58 (1.31)	30%
Moderate-Severe	8	1027	13.75 (2.92)	20%

3.4 DepressionAnxiety Covariance

Pearson correlations revealed strong intercorrelations among the clinical scales: **PHQ-9 & GAD-7**: $r = .579$, $p < .001$; **PHQ-9 & STAI-T**: $r = .642$, $p < .001$; **GAD-7 & STAI-T**: $r = .697$, $p < .001$. These are consistent with the well-established comorbidity between depression and anxiety in the literature [10] and broadly similar to those reported by Srivastava and Lahane [4] ($r = .75$ in their sample).

Descriptive comparison across depression groups shows that GAD-7 and STAI-T scores increase with depression severity (Minimal: GAD $M = 2.95$, STAI $M = 37.75$; Moderate-Severe: GAD $M = 8.38$, STAI $M = 57.62$). Kruskal-Wallis tests confirmed that GAD-7 ($H = 12.81$, $p = .002$) and STAI-T ($H = 10.63$, $p = .005$) both differed significantly across depression groups, meaning the groups are not matched on anxiety. This presents a potential confound for any analysis comparing head movement across depression groups, since participants with higher depression also have higher anxiety. Figure 2 presents the scatter plots and correlation heatmap.

3.5 Per-Video Subjective Ratings

Figure 3 presents mean valence, arousal, and immersion across all five videos. **V4 (The Nun)** produced the lowest valence ($M = 3.67$, $SD = 1.85$) and highest arousal ($M = 5.95$, $SD = 1.68$), while **V5 (Tahiti Surf)** yielded the highest valence ($M = 7.12$, $SD = 1.68$) and highest arousal overall. These patterns are consistent with the intended emotional content of each video and **confirm that the stimuli elicited meaningfully different affective responses across participants**.

Notably, V4 also showed the **lowest immersion score** ($M = 19.85$) despite being substantially longer than V1-V3. This **suggests that high arousal does not necessarily translate into greater sense of presence**; the threatening horror content may have produced discomfort-driven withdrawal rather than engagement.

3.6 Mood Change: PANAS Pre- vs. Post-VR

A paired-samples t-test revealed a **significant decrease in positive affect** following the VR session ($M_{\text{pre}} = 34.52$, $M_{\text{post}} = 32.15$; $t(39) = 2.09$, $p = .043$, Cohen's $d = 0.27$). **Negative affect did not change significantly** ($t(39) = -1.06$, $p = .294$). Figure 4 shows mood trends broken down by depression group; the **Moderate-Severe group showed the numerically largest decline in positive affect.**

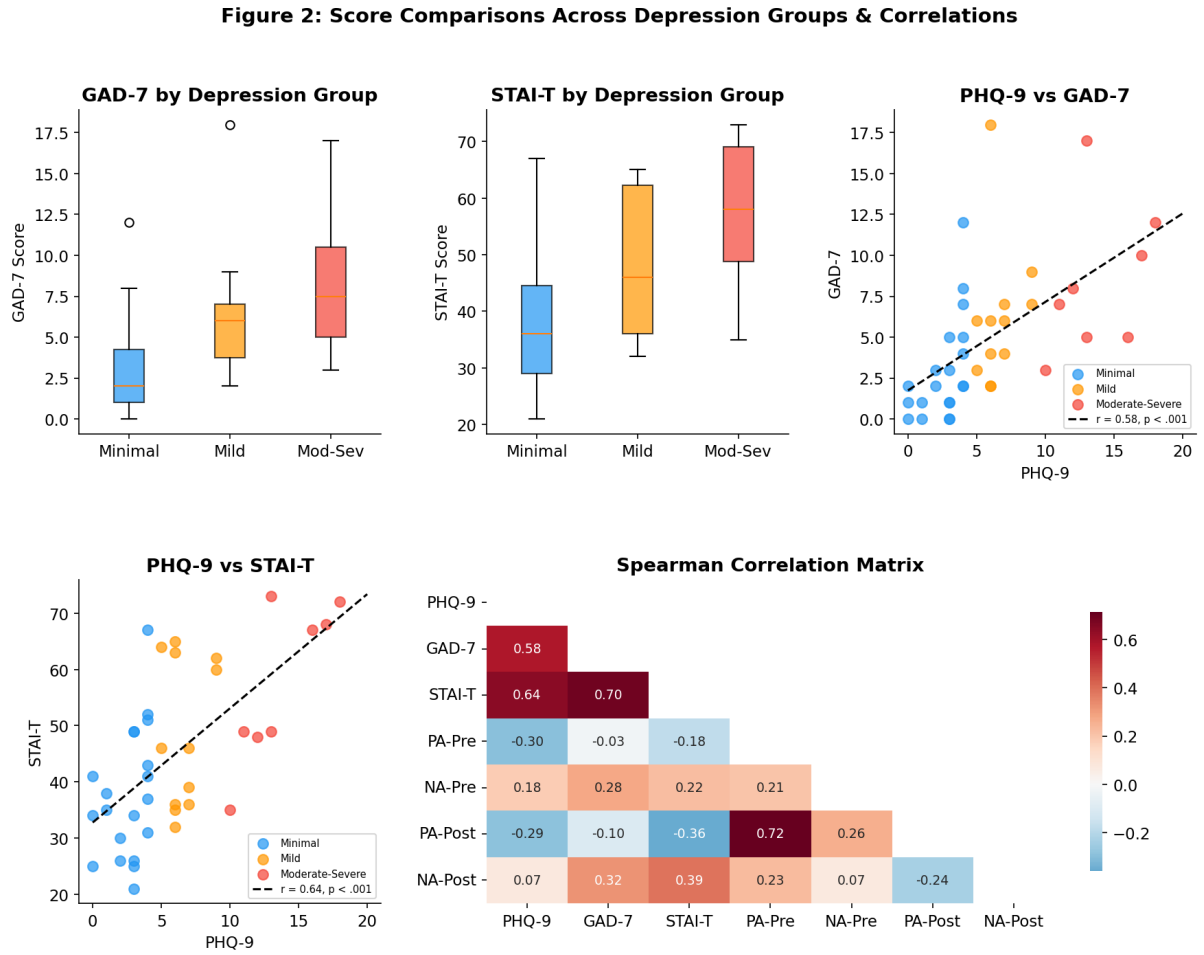

Figure 2: (Left) Boxplots of GAD-7 and STAI-T by depression group. (Right) Scatter plots of PHQ-9 vs. GAD-7 and PHQ-9 vs. STAI-T; Pearson correlation heatmap for all key variables.

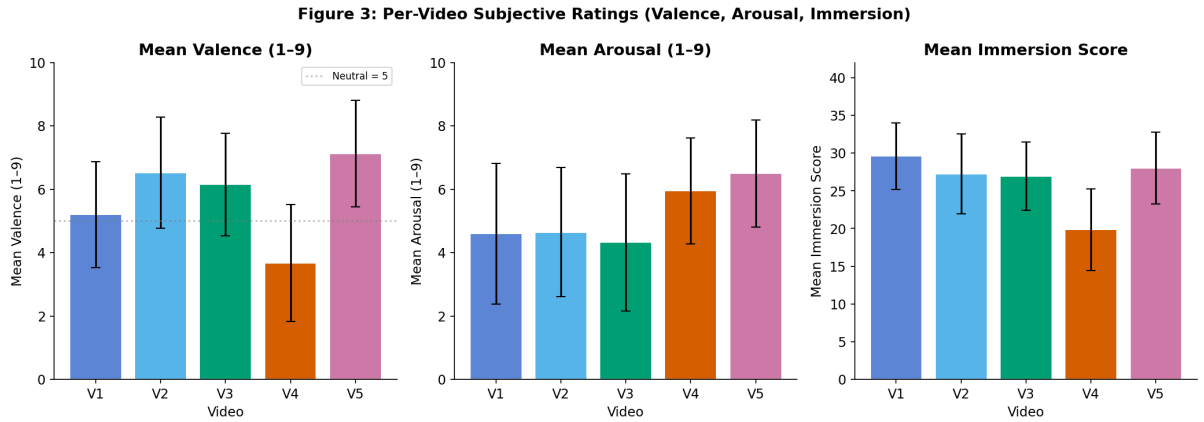

Figure 3: Mean valence, arousal, and immersion per video (error bars = ±1 SD). V1 = Abandoned Buildings, V2 = Beach, V3 = Campus, V4 = Horror, V5 = Surf.

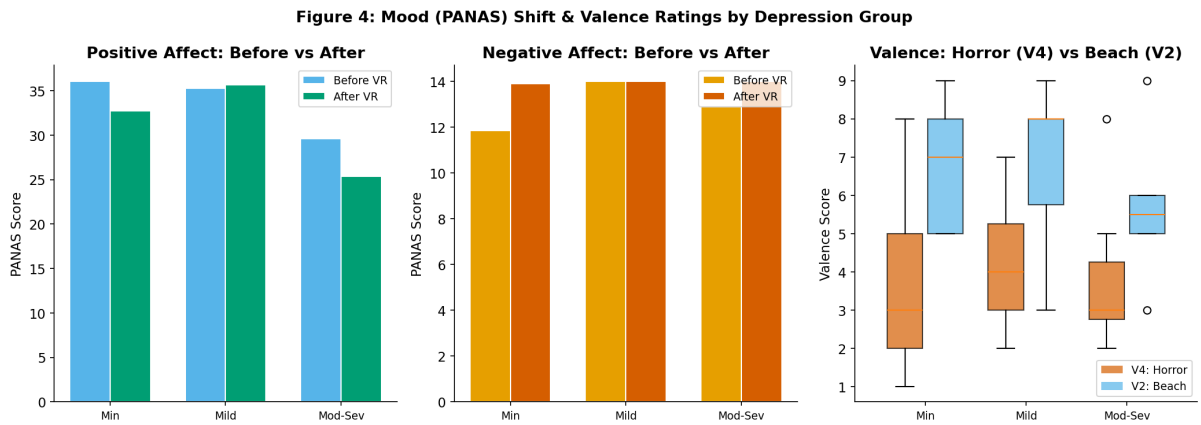

Figure 4: PANAS positive and negative affect before vs. after VR by depression group (left two panels); valence ratings for Horror (V4) vs. Beach (V2) by group (right panel).

3.7 Head-Tracking: Descriptive Overview of Yaw Scanning Speed

Table 3 presents descriptive statistics for mean yaw angular speed (deg/s) per video across all participants. The most striking finding is that **V4 (Horror) produced dramatically lower scanning speed** ($M = 17.92$ deg/s) compared to all other videos (range: 25.9531.19 deg/s). This is consistent with threat-induced freezing behaviour, where the perceived danger of the horror content suppressed exploratory head movement despite the highest reported arousal.

Table 3: Descriptive statistics for mean yaw scanning speed (deg/s) per video ($N = 40$).

Video	Name	Mean	SD	Min	Max	Duration (s)
V1	Abandoned Buildings	31.19	8.77	9.41	47.32	≈60
V2	Evening at Beach	28.14	7.73	6.48	45.27	≈60
V3	Campus	29.73	9.39	12.56	47.68	≈60
V4	The Nun (Horror)	17.92	8.28	4.52	34.23	≈180
V5	Tahiti Surf	25.95	9.64	10.23	48.69	≈180

Figure 5 shows yaw speed distributions per video and group-level comparisons. Kruskal-Wallis tests found no statistically significant differences in scanning speed across depression groups for any video (all $p > .30$; e.g., V1: $H = 0.83$, $p = .66$; V3: $H = 1.04$, $p = .59$; V5: $H = 2.14$, $p = .34$). However, the directional trend is noteworthy: for V1, V3, and V5, the Moderate-Severe group showed numerically lower yaw speed than the Minimal group (V3: $M_{\text{Minimal}} = 31.32$, $M_{\text{Mod-Sev}} = 27.07$; V5: $M_{\text{Minimal}} = 28.31$, $M_{\text{Mod-Sev}} = 24.16$), consistent with the direction predicted by Srivastava and Lahane [4]. However, the group differences are modest and overlap substantially in their distributions. Spearman correlations between PHQ-9 and yaw **speed were small and non-significant across all videos** (all $|\rho| < .12$, $p > .48$).

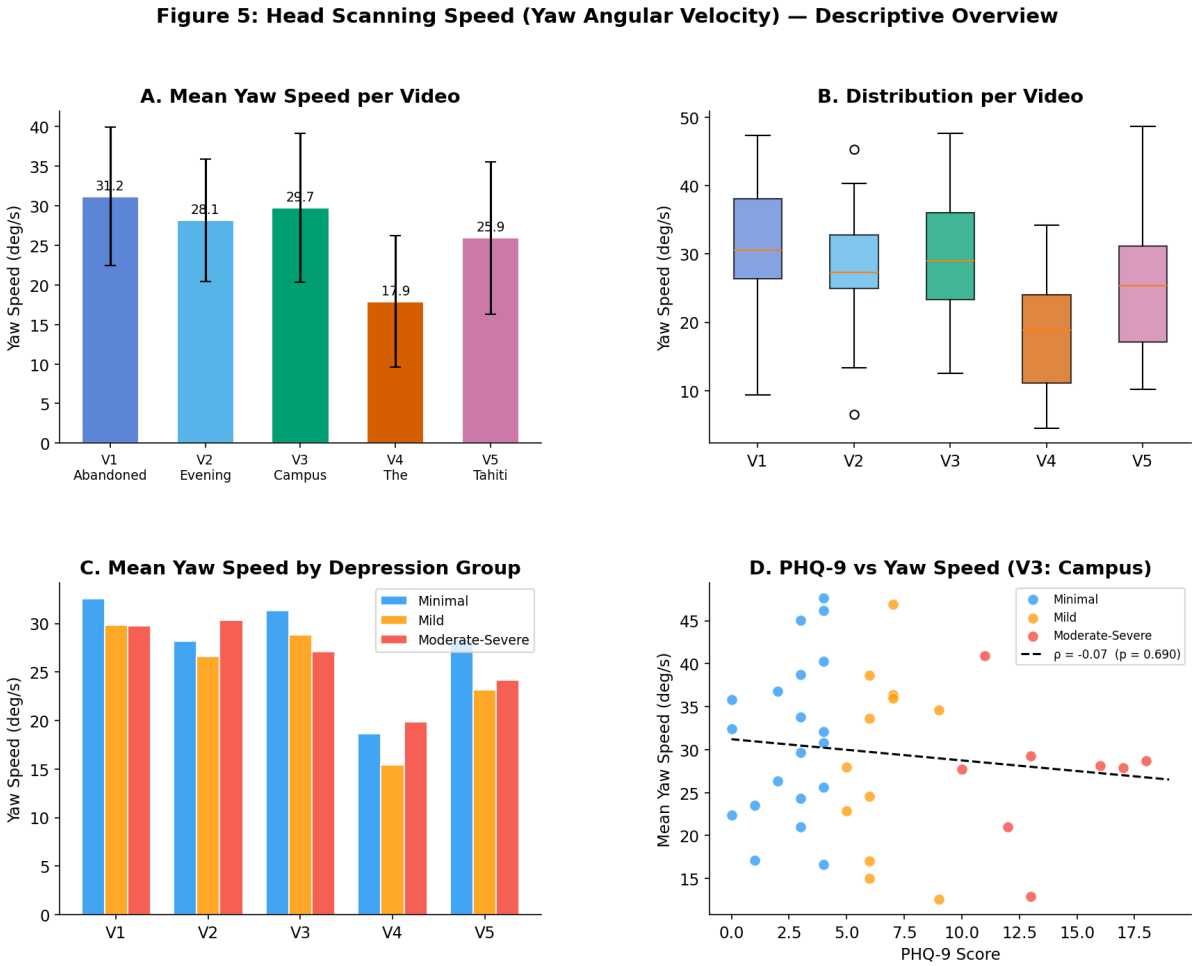

Figure 5: Head scanning speed (yaw angular velocity, deg/s). A: Mean per video with SD error bars. B: Distribution boxplots per video. C: Group means by depression group across videos. D: PHQ-9 vs. yaw speed scatter for V3 with trend line and Spearman ρ.

4 Conclusion

This preliminary analysis established the following findings. First, PHQ-9, GAD-7, and STAI-T scores all **deviated from normality** (Shapiro-Wilk). Second, **depression** and **anxiety** measures were **strongly intercorrelated** (r = .58-.64), a design feature common to studies with naturally-occurring depression groups. Third, different 360° videos reliably **elicited** distinct **emotional responses**: the horror video (V4) **produced** the **lowest valence and lowest immersion despite highest arousal**, and the lowest head scanning speed (M = 17.92 deg/s), compared to $M \approx$ 25-31 deg/s for the other four videos. Fourth, the VR session produced a small but **significant decline in positive affect** ($t(39) = 2.09$, $p = .043$, $d = 0.27$), with the **Moderate-Severe group showing the largest drop.**

On the core hypothesis that depression severity predicts reduced head scanning speed the data show the correct directional trend for three of five videos, but Kruskal-Wallis tests do not reach significance (all $p > .30$) and Spearman correlations between PHQ-9 and yaw speed are near zero across all videos. A notable complication is that depression groups also differ significantly on anxiety (GAD-7 and STAI-T; Kruskal-Wallis $p < .01$), meaning any group difference in scanning speed could reflect anxiety rather than depression.

Future Work

Report 2 will build on these preliminary findings by applying further inferential statistical methods to test whether depression severity is associated with reduced head scanning speed, accounting for the confounding role of anxiety. The additional measures collected in this study PANAS affect scores and per-video immersion ratings will be examined for their relationship with head movement behaviour. Results will be discussed in terms of how closely they replicate the findings of Srivastava and Lahane [4].

Author Contributions

All three authors contributed equally to this work. **Rachakonda Hrithik Sagar** led the data processing, descriptive analysis pipeline and report writing. **Aryan Jain** led the inferential statistical analyses and report writing. **Vidhi Rathore** led the head-tracking analysis. All authors participated in the VR experiment, read and discussed the reference literature, interpreted the results, and reviewed the final manuscript.

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